BASE CHANGE ALONG A CLOSED IMMERSION UNDER TOR INDEPENDENCE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The pushforward along a closed immersion is exact, so the derived pushforward is computed on any representative, and the base change map of the cartesian square can be evaluated on stalks. used at proof lines 5567-5575. Reduces the invertibility of the functor identity to a stalkwise tensor statement.
- The algebraic base change comparison for a compatible pair of local ring maps is an isomorphism exactly under the stated tor independence. invoked at proof lines 5580-5584. Identifies the displayed stalkwise tensor map with the isomorphism supplied by the More on Algebra lemma.

Source reference. Original lemma 0661: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 5580-5584 together with tor independence to conclude the stalkwise map is an isomorphism; the stalk identification it is applied to is inside the counted span.

Lemma 0.2. *Consider a cartesian diagram of schemes*

$$\begin{array}{ccc} Z' & \xrightarrow{i'} & X' \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{i} & X \end{array}$$

where i is a closed immersion. If Z and X' are tor independent over X, then $Ri'_* \circ Lg^* = Lf^* \circ Ri_*$ as functors $D(\mathcal{O}_Z) \to D(\mathcal{O}_{X'})$.

Proof. Note that the lemma is supposed to hold for all $K \in D(\mathcal{O}_Z)$. Observe that i_* and i'_* are exact functors and hence Ri_* and Ri'_* are computed by applying i_* and i'_* to any representatives. Thus the base change map

$$Lf^*(Ri_*(K)) \longrightarrow Ri'_*(Lg^*(K))$$

on stalks at a point $z' \in Z'$ with image $z \in Z$ is given by

$$K_z \otimes^{\mathbf{L}}_{\mathcal{O}_{X,z}} \mathcal{O}_{X',z'} \longrightarrow K_z \otimes^{\mathbf{L}}_{\mathcal{O}_{Z,z}} \mathcal{O}_{Z',z'}$$

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This map is an isomorphism by More on Algebra, Lemma 0661 and the assumed tor independence. □